Roll No. ....

TEE–304

B. TECH. (EE) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

ELECTROMAGNETIC FIELDS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain cross and dot product of two vectors. Also, explain the method to find direction cosine of any vector. (CO1)

OR

(b) Discuss Spherical Coordinate Systems. Also, find the relation between unit vectors in Cartesian and spherical coordinate systems. (CO1)

2. (a) Explain Curl, Divergence and Gradient of the vector field. Also discuss the physical interpretation of curl and divergence. (CO1)

OR

(b) Prove the following : (CO1)

(i) $\nabla \cdot \vec{r} = 3$

(ii) $\nabla \times \vec{r} = 3$, where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

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3. (a) Discuss Circular Cylindrical Coordinate Systems. Also, find the relation between unit vectors in Cartesian and cylindrical coordinate systems. (CO1)

OR

(b) Find the location of the point (2, –1, 3) in cylindrical and spherical coordinate systems. (CO1)

4. (a) Discuss Coulomb's law, electrical field intensity and electric field potential. (CO2)

OR

(b) The concentrated charge of 0.25 μC are placed at the vertices of an equilateral triangle whose side is 100 mm. Determine the magnitude and direction of the resultant force on one charge due to the other charges. (CO3)

5. (a) Derive the expression of electric field due to finite line of charge. (CO2)

OR

(b) The potential distribution in a given region of space is of the form $V = 10y^3 + 20x^2$. Obtain the value of $\vec{E}$ at (10, 0) and (14, 16). (CO3)